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An analysis of open-basin lake deposits on Mars: Evidence for the nature of associated lacustrine deposits and post-lacustrine modification processes

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ABSTRACT

A large number of candidate open-basin lakes (low-lying regions with both inlet valleys and an outlet valley) have been identified and mapped on Mars and are fed by valley network systems that were active near the Noachian–Hesperian boundary. The nature of processes that modified the open-basin lake interiors subsequent to lacustrine activity, and how frequently sedimentary deposits related to lacustrine activity remain exposed, has not been extensively examined. An analysis of 226 open-basin lakes was undertaken to identify evidence for: (1) exposed deposits of possible lacustrine origin and (2) post-lacustrine-activity processes that may have modified or resurfaced open-basin lakes. Spectroscopic data from the Compact Reconnaissance Imaging Spectrometer for Mars (CRISM) instrument were analyzed over identified exposed open-basin lake deposits to assess the mineralogy of these deposits. Particular attention was paid to the possible detection of any component of aqueous alteration minerals (e.g. phyllosilicates, hydrated silica, zeolites) or evaporites (e.g. carbonates, sulfates, chlorides) associated with these exposed deposits. The aim of this paper is to act as a broad survey and cataloguing of the types of lacustrine and post-lacustrine deposits that are present within these 226 paleolake basins. Results of the morphologic classification indicate that 79 open-basin lakes (~35% of the population) contain exposed deposits of possible lacustrine origin, identified on the basis of fan/delta deposits, layered deposits and/or exposed floor material of apparent lacustrine origin. Additionally, all 226 open-basin lakes examined appear to have been at least partially resurfaced subsequent to their formation by several processes, including volcanism, glacial and periglacial activity, impact cratering and aeolian activity. Results from the analysis of CRISM data show that only 10 (~29% of the 34 deposits with CRISM coverage) of the exposed open-basin lake deposits contain positively identified aqueous alteration minerals, with one deposit also containing evaporites. The identified hydrated and evaporite minerals include Fe/Mg-smectite, kaolinite, hydrated silica and carbonate, with Fe/Mg-smectite the most commonly identified mineral. These results indicate that hydrated and evaporite minerals are not as commonly associated with lacustrine deposits on Mars as they are on Earth. This suggests *in situ* alteration and mineral precipitation, a common source of such minerals in terrestrial lakes, was not a major process occurring in these paleolacustrine systems, and that the observed minerals are likely to be present as transported material within the lacustrine deposits. The lack of widespread *in situ* alteration also suggests that either the water chemistry in these paleolake systems was not conducive to aqueous alteration and mineral precipitation, or that the open-basin lake systems were relatively short-lived.

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1. Introduction

Candidate paleolake basins have long been observed on Mars (e.g. Goldspiel and Squyres, 1991; De Hon, 1992; Forsythe and Zimbelman, 1995; Cabrol and Grin, 1999, 2001) based on their distinct morphology. Since their initial discovery, several workers have compiled extensive and thorough catalogues of these features (e.g. De Hon, 1992; Cabrol and Grin, 1999, 2001; Fassett and Head, 2008a), dividing martian paleolakes into two major categories: closed-basin lakes and open-basin lakes (Cabrol and Grin, 1999; Fassett and Head, 2008a). Closed-basin lakes have inlet valleys but lack outlets, and they are inferred to be paleolakes due to the observed morphology of the surrounding terrain (i.e. inlet valleys) and associated deposits (Cabrol and Grin, 1999). Open-basin lakes have both observed inlet valleys and an outlet valley (Cabrol and Grin, 1999; Fassett and Head, 2008a). The presence of both inlet valleys and an outlet valley means that water within the basin must have ponded to approximately the level of the surface adjacent to the outlet valley head before breaching and overflowing

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the basin, requiring a period of sustained fluvial activity on the surface of Mars (Fassett and Head, 2008a).

Over 200 open-basin lakes have been mapped in the ancient Noachian and Hesperian highlands across much of the planet (Fassett and Head, 2008a). These open lacustrine systems have been interpreted to have been active in a distinct period in martian history, with the majority of fluvial activity within the valley networks that feed them ceasing at or near the Noachian–Hesperian boundary (Irwin et al., 2005; Fassett and Head, 2008b); however, it is possible that a very small percentage of these basins were active at a later period in martian history. Therefore, the currently exposed deposits associated with these open-basin lakes have had a ~3.7 Gyr history that has been largely devoid of fluvial activity (Fassett and Head, 2008a, 2008b). During this extensive period of time, the basins that hosted these paleolakes have been subject to a variety of geologic processes that have acted to resurface, bury, and exhume the open-basin lakes and any exposed deposits associated with them.

In this investigation we conduct a comprehensive analysis of the morphologies and mineralogies associated with the entire catalogue of 210 open-basin lakes from Fassett and Head (2008a) in addition to 16 open-basin lakes added to the catalogue subsequently. The aim of this work is to conduct an analysis of the presence, type and composition of exposed sedimentary deposits associated with the open-basin lakes as well as the classification of any identifiable resurfacing units modifying or covering the basins. The analyses presented here reflect this scope of a broad survey of deposits associated with open-basin lakes, and are not intended to act as a detailed morphologic, sedimentologic or mineralogic study of any one deposit or basin alone, as have been done by many previous workers (e.g. Grin and Cabrol, 1997; Ori et al., 2000; Irwin et al., 2002; Fassett and Head, 2005; Mangold and Ansan, 2006; Ehlmann et al., 2008a, 2008b; Dehouck et al., 2010; Ansan et al., 2011; Buhler et al., 2011).

A major goal in assessing the exposed open-basin lake deposit's composition is to determine if there is any common association with aqueous alteration minerals (e.g. phyllosilicates, hydrated silica, zeolites) and/or evaporites (e.g. carbonates, chlorides, sulfates), as is commonly seen for lacustrine deposits on Earth (Jones and Bowser, 1978; Blatt et al., 1980; Leeder, 1999; Wetzel, 2001). In terrestrial systems, aqueous alteration minerals occur in lacustrine sediments as both transported detritus and as authigenic material formed through lithification and diagenesis (Moore, 1961; Muller and Quakernaat, 1969; Singer et al., 1972; Jones and Bowser, 1978; Blatt et al., 1980; Jones, 1986; Hay et al., 1991; Hillier, 1993), whereas evaporites typically only form as authigenic components of the sediment from mineral precipitation in the water column and through diagenesis (Brunskill, 1969; Eugster and Hardie, 1975, 1978; Jones and Bowser, 1978; Kelts and Hsu, 1978; Strong and Eadie, 1978; Wetzel, 2001). Aqueous alteration minerals commonly comprise a major component (>~15–25 wt.%) of lacustrine sediments from large, open lake systems, while evaporites (primarily carbonate) are present at variable levels depending on the chemistry of the lake water (e.g. Moore, 1961; Thomas, 1969; Muller and Quakernaat, 1969; Singer et al., 1972; Thomas et al., 1972, 1973; Jones and Bowser, 1978; Blatt et al., 1980; Hillier, 1993). The overall aim of this study is to further characterize the catalogue of open-basin lakes observed across the surface of Mars to help understand both the period of lacustrine activity within the open-basin lakes and the post-fluvial-activity history of these basins.

2. Datasets used and analysis

In order to examine the morphology of the open-basin lakes, a combination of ~6 m/pixel images from the Context Camera (CTX) instrument aboard the Mars Reconnaissance Orbiter (MRO) spacecraft (Malin et al., 2007), ~18 m/pixel images from the visible camera of the Thermal Emission Imaging System (THEMIS) instrument aboard the Mars Odyssey spacecraft (Christensen et al., 2004) and <50 m/pixel images from the High Resolution Stereo Camera (HRSC) instrument aboard the Mars Express spacecraft (Neukum et al., 2004) were used. Additionally, Mars Orbiter Laser Altimeter (MOLA) gridded topography (Smith et al., 2001) and HRSC stereo topography (Neukum et al., 2004) were used to further characterize the morphology of the open-basin lakes and their associated deposits.

Each open-basin lake in the Fassett and Head (2008a) catalogue was examined at a variety of scales and was categorized based on two main aspects of their morphology: (1) presence or absence of exposed deposits that appear to be lacustrine in nature and (2) characteristics of any process or processes that have been involved in resurfacing the open-basin lakes. It should be noted that the classification of having undergone resurfacing and having exposed deposits of possible lacustrine origin are not mutually exclusive, as an open-basin lake can be partially resurfaced while still containing exposed sedimentary deposits. The criteria used for classification of the presence of exposed sedimentary deposits and resurfacing processes are discussed in Sections 2.1 and 2.2.

During this analysis, we also reassessed the evidence for paleolake activity associated with each of the 226 open-basin lakes. Images over each open-basin lake were examined for discernable evidence of an outlet valley that is hanging with respect to a contour level that defines a local topographic low and does not head at a drainage divide. The incision of such valleys cannot be explained by localized precipitation or run-off, and thus requires a standing body of water, at some point in time, that has been breached and supplied the fluvial activity necessary for valley incision (Fassett and Head, 2008a). Each open-basin lake was assigned one of three degrees of confidence for paleolake activity associated with the basin: certain, probable and possible. Certain indicates that there is an extremely high degree of confidence that there was a paleolake in the analyzed basin, probable indicates that it is highly likely that there was a paleolake in the analyzed basin and possible indicates that it is likely there was a paleolake in the analyzed basin, although there is some degree of uncertainty.

Exposed open-basin lake deposit mineralogies were assessed using spectral data from the Compact Reconnaissance Imaging Spectrometer for Mars (CRISM) instrument aboard the MRO spacecraft (Murchie et al., 2007). Each exposed open-basin lake deposit was studied using full resolution, FRT, (~18 m/pixel) and half resolution, HRL and HRS, (~36 m/pixel) targeted observations where such data are available, as these data have been shown to have a sufficiently high resolution to identify the composition of exposed sedimentary deposits (e.g. Ehlmann et al., 2008a, 2008b, 2009; Mustard et al., 2008; Dehouck et al., 2010; Ansan et al., 2011). This assessment was performed using both spectral parameter maps, which identify key spectral absorptions caused by components of the mineral structure such as bound water and hydroxyl molecules (Pelkey et al., 2007; Ehlmann et al., 2009), as well as validation of mineral detections identified through parameters using detailed spectral analysis.

2.1. Exposed open-basin lake deposits

Open-basin lakes were examined for three primary types of exposed sedimentary deposits: fan/delta deposits, layered deposits and exposed floor material. The criteria used to identify these three types of deposits are detailed below.

2.1.1. Deltas

Images of open-basin lake inlet valleys were analyzed to document any evidence of sediment deposition in the form of a fan or

delta deposit (Fig. 1). These types of deposits have been observed in numerous other open-basin lakes (e.g. Cabrol and Grin, 1999; Fassett and Head, 2005; Irwin et al., 2005; Mangold and Ansan, 2006; Di Achille and Hynek, 2010) and provide strong evidence for fluvial activity. The morphology of these fan deposits is highly variable across the surface of Mars, and while several workers have documented these variations (e.g. Irwin et al., 2005; Di Achille and Hynek, 2010), here we simply note the presence or absence of delta deposits within the open-basin lakes. It is important to note that while relatively pristine delta deposits have been observed in a few open-basin lakes (e.g. Fassett and Head, 2005; Irwin et al., 2005; Mangold and Ansan, 2006) (Fig. 1A and D), the majority of the delta deposits we identify here have been heavily eroded and/or modified (Fig. 1B, C, E and F).

2.1.2. Layered deposits

Layered sedimentary deposits have been observed in a number of locations across Mars with high-resolution imagery (e.g. Malin and Edgett, 2000) including in several open-basin lakes (e.g. Cabrol and Grin, 1999; Wilson et al., 2007). In this study we analyzed images of each open-basin lake and noted whether or not they contain such exposed between layered deposits (Fig. 2). In order to distinguish between layered deposits with a probable lacustrine origin and those deposited through glacial (e.g. Head et al., 2003) or aeolian (e.g. Lewis et al., 2008) processes, we classified basins as showing evidence for layering only if: (1) the layered deposits are situated topographically below the outlet level of the basin based on MOLA gridded data (Smith et al., 2001) and/or HRSC stereo data (Neukum et al., 2004); (2) the layered deposits are stratigraphically below any resurfacing unit; (3) layers are contiguous across the basin where separate outcrops are visible; and (4) outcropping layers are wholly confined to the associated open-basin lake.

2.1.3. Exposed floor material

A final type of exposed deposit of possible lacustrine origin identified is exposed floor material (Fig. 3). These deposits are typically present as light-toned, knobby, eroded terrain on the floor of open-basin lakes, which are either embayed by or exposed beneath any resurfacing unit present (Fig. 3). While it is possible that these deposits are not lacustrine in origin, based on several lines of evidence we conclude that lacustrine deposition is the most likely source for these deposits. First, the deposits all lie stratigraphically below any resurfacing unit and are commonly embayed by these resurfacing units (Fig. 3). Additionally, these deposits are typically light- to intermediate-toned and often form mesas, typical of massive sedimentary deposits observed across the surface of Mars (Malin and Edgett, 2000). Furthermore, a period of extensive deposition is implied by the abnormally low depth/diameter relationships (Strom et al., 1992) for those basins defined by ancient impact craters. That this deposition is lacustrine is favored by the confinement of the deposits to paleolake basins. Although we favor the hypothesis of lacustrine deposition for these exposed floor

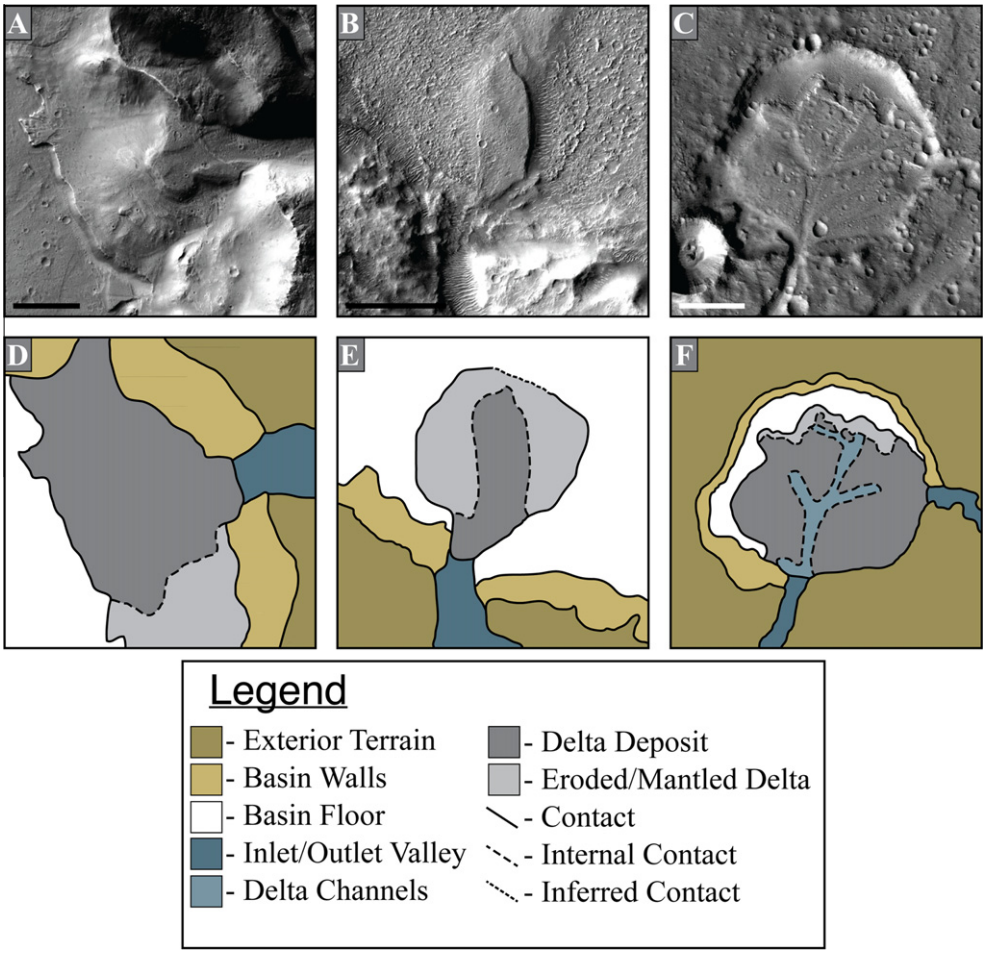


Fig. 1. Examples of various types of delta deposits observed in open-basin lakes. Scale bars are 2 km. (A) Pristine Gilbert-type delta at 39.16°S, 103.12°W (Mangold and Ansan, 2006). CTX image P15_006798_1405_XI_39S103 W. (B) Partially eroded delta at 22.38°S, 23.68°W. CTX image B17_016170_1572_XI_22S023 W. (C) Partially eroded delta at 8.53°N, 48.01°W. CTX image P06_003407_1872_XN_07N047 W (Harrison and Grimm, 2005; Hauber et al., 2009). (D–F) Geologic sketch maps of the delta deposits in parts (A–C) respectively. Legend at bottom indicates contact and unit types.

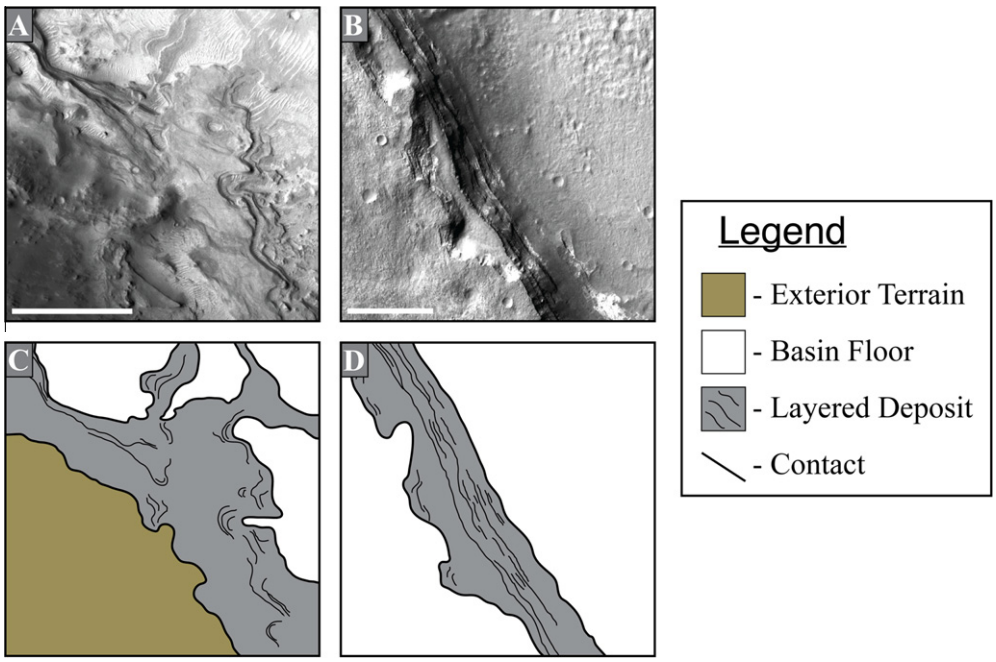


Fig. 2. Examples of various types of layered deposits observed in open-basin lakes. Scale bars are 2 km. (A) Layered deposit at 26.98°N, 74.17°E. CTX image P17_007490_2095_XN_29N286 W. (B) Layered deposit in Terby Crater, 28.25°S, 73.68°E (Wilson et al., 2007). CTX image P15_007042_1519_XI_28S286 W. (C and D) Geologic sketch maps of the layered deposits in parts (A and B) respectively. Legend at right indicates contact and unit types.

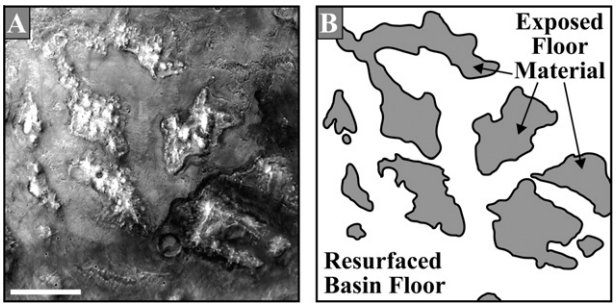


Fig. 3. (A) Exposed floor material of likely lacustrine origin at 4.37°S, 1.71°W. Scale bar is 2 km. Note the light-toned, knobby appearance of the exposed floor material, which is being embayed by the darker, volcanic resurfacing unit on the basin floor. CTX image P13_006214_1765_XN_03S001 W. (B) Geologic sketch map of the exposed floor material in part (A).

materials, it is possible that some portion of these deposits are unrelated to lacustrine activity. In identifying exposed floor materials, care was taken to ensure that exposed central peak and/or peak ring material was not confused with exposed floor material in open-basin lakes defined by ancient impact craters.

2.2. Resurfacing units

In addition to identifying the presence or absence of exposed sedimentary deposits, each open-basin lake was classified as to the presence of modifying and resurfacing units that were deposited subsequent to the lacustrine activity within the basin. The type of resurfacing process that has most recently affected the basin was also identified where such an identification and classification were possible, with the main processes described below.

2.2.1. Volcanic resurfacing

Volcanically resurfaced open-basin lakes were identified based on the presence of several distinct morphologies that indicate resurfacing by volcanic flows (Fig. 4). First, volcanically resurfaced open-basin lakes contain smooth floor deposits with high crater retention, especially at small crater sizes, suggesting a competent material (Fassett and Head, 2008a). Additionally, these basins commonly have wrinkle ridges on their floors, a morphology regularly observed in volcanic plains across the surface of Mars (Watters, 1991) (Fig. 4). Furthermore, these basins typically have lobate margins that appear to embay the open-basin lake perimeter (typically crater walls) and any older floor material or exposed sedimentary deposits that may be present (Figs. 3A and 4). Finally, many of these basins exhibit a "moat" morphology at their edges, which is typical of volcanically flooded basins and is thought to be caused by the subsidence of volcanic fill material (e.g. Leverington and Maxwell, 2004) or the flow stopping at the base of a rise at the margin of the crater wall (Fig. 4A).

2.2.2. Glacial resurfacing

Glacially modified and resurfaced craters, such as those containing concentric crater fill (Levy et al., 2010; Dickson et al., 2010, 2011), show flow patterns, features and structures that can be used as guides for the identification of glacial resurfacing processes. Open-basin lakes classified as being glacially resurfaced exhibit such features, typically lobate floor texture and lobate ridges (Fig. 5), indicative of material deposition by glacial processes (e.g. Head et al., 2008). These ridges are distinguished from paleoshorelines based on their lobate morphology that is not constrained by or following pre-existing topography, as would be expected for shorelines (Parker et al., 1993; Head et al., 1998). Glacially resurfaced open-basin lakes also typically exhibit ring-mold craters, which provide evidence for current or past subsurface ice deposits (Mangold, 2003; Kress and Head, 2008) (Fig. 5).

2.2.3. Resurfacing of unknown origin

Several open-basin lakes show evidence for resurfacing and/or modification across parts of their basin floors with no single process clearly evident as the cause for resurfacing (Fig. 6). The evidence for resurfacing in these basins is: (1) the heavy erosion

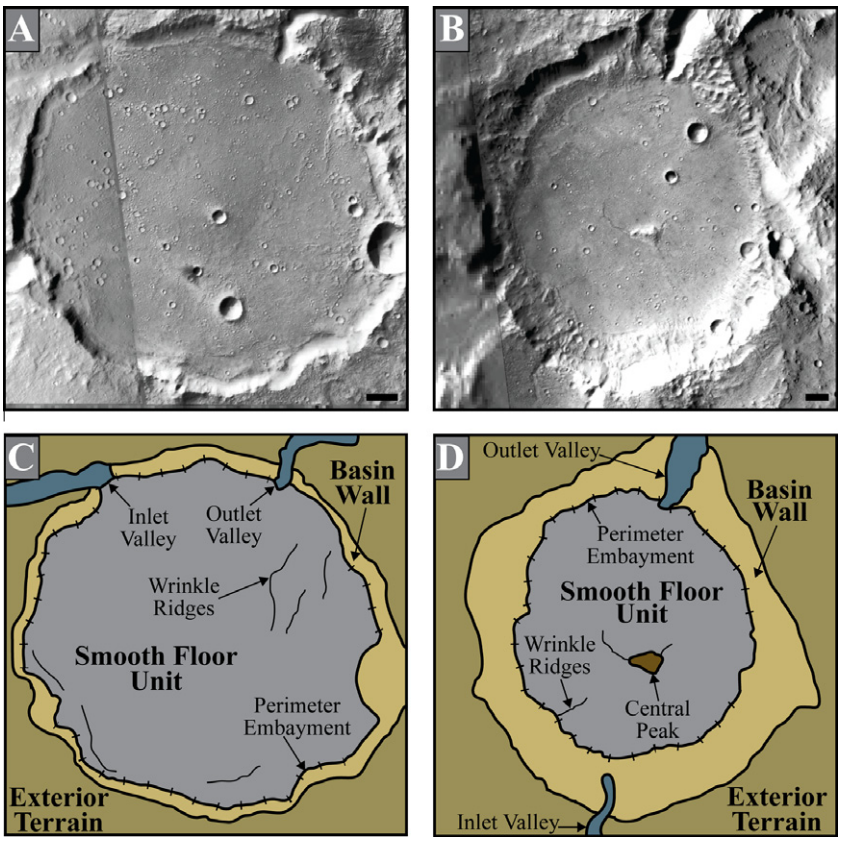


Fig. 4. Examples of volcanically resurfaced open-basin lakes. Note the smooth plains appearance on the basin floor and high crater retention. Scale bars are 2 km. (A) Volcanically resurfaced open-basin lake at 12.44°S, 157.12°E (Fassett and Head, 2008a). CTX images G04_019750_1675_XN_12S203 W and P19_008555_1676_XN_12S202 W overlain on THEMIS visible mosaic. (B) Volcanically resurfaced open-basin lake at 11.74°S, 144.06°E (Cabrol and Grin, 1999; Fassett and Head, 2008a). Mosaic of CTX images B18_016599_1686_XN_11S216 W and G05_020001_1689_XN_11S215 W overlain on THEMIS visible mosaic. (C and D) Geologic sketch maps of the volcanically resurfaced open-basin lakes in parts (A and B) respectively. Wrinkle ridges and basin perimeter embayment are indicated.

and denudation of the inlet and outlet valleys, which we hypothesize has also affected the basin floor; (2) a consistent surface texture across the basin floor and the surrounding terrain, suggesting a continuous unit; and (3) impact craters within the basin that are visibly buried by a mantling unit (Fig. 6). While all of these lines of evidence point toward post-fluvial-activity resurfacing and modification, conclusively determining the process that resurfaced these basins is difficult in the reconnaissance mode and requires careful examination and comprehensive analysis of each individual basin. Candidate regional resurfacing processes include: (1) emplacement of impact ejecta (Scott and Tanaka, 1986; Greeley and Guest, 1987); (2) deposition of dust or other aeolian mantling materials (Scott and Tanaka, 1986; Greeley and Guest, 1987; Tanaka, 2000); (3) climate-related mantling by ice-rich deposits (e.g. Mustard et al., 2001; Head et al., 2003); or (4) aeolian erosion of friable surface materials (e.g. McCauley, 1973; Arvidson et al., 2003; Kerber and Head, 2010).

3. Results

On the basis of our morphologic examination of 226 open-basin lakes (Fassett and Head, 2008a), 79 (~35%) contain exposed deposits of possible lacustrine origin, while 147 (~65%) show no discernable evidence for such deposits on the surface (Fig. 7, Table 1). Additionally, the entire population of open-basin lakes appears at least partially resurfaced. The most prominent identifiable resurfacing unit is of volcanic origin, with 96 open-basin lakes (~42%) identified as volcanically resurfaced, while only 21 (~9%) are glacially resurfaced and 109 (~49%) appear resurfaced, but do not have a clear specific process identified for the resurfacing (Fig. 8, Table 1). From the results of our morphologic survey (Table 1), we also find that of the 226 basins examined, 210 (~93%) have a certain degree of confidence of paleolake activity, 9 (~4%) have a probable degree of confidence of paleolake activity and 7 (~3%) have a possible degree of confidence of paleolake activity.

Based on the results of the analysis of CRISM targeted observations over exposed open-basin lake deposits, we find that of the 79 open-basin lakes with observed sedimentary deposits, 34 have targeted CRISM observations that cover these deposits (Table 2). Of these 34 open-basin lake deposits, we find that 10 (Table 3) have spectral signatures consistent with the presence of aqueous alteration minerals, with one of these ten also containing the only observed spectral signature consistent with evaporite minerals (Table 3). The identified minerals are contained within the exposed sedimentary deposits (Fig. 9A–F) and do not appear present in the surrounding terrain. The remaining analyzed open-basin lake deposits do not show a strong spectral diversity from their surrounding terrain (Fig. 9G and H), and give no strong indication for unique mineral detections, hydrated or otherwise.

Identified aqueous alteration minerals include Fe/Mg-smectite, kaolinite and hydrated silica, and the one identified evaporite mineral is carbonate (Fig. 10). The identification of Fe/Mg-smectite is based on prominent absorptions at 1.4, 1.9 and ~2.3 μm (Fig. 10). Specifically, the 1.4 μm absorption is due to the overtone of a fundamental absorption from structural OH, the 1.9 μm absorption is due to a combination of the fundamental absorption from OH stretch and H_2O bend, and the ~2.3 μm absorption is due to a combination tone of a fundamental absorption of the metal–OH bond (Clark et al., 1990; Frost et al., 2002), with the precise

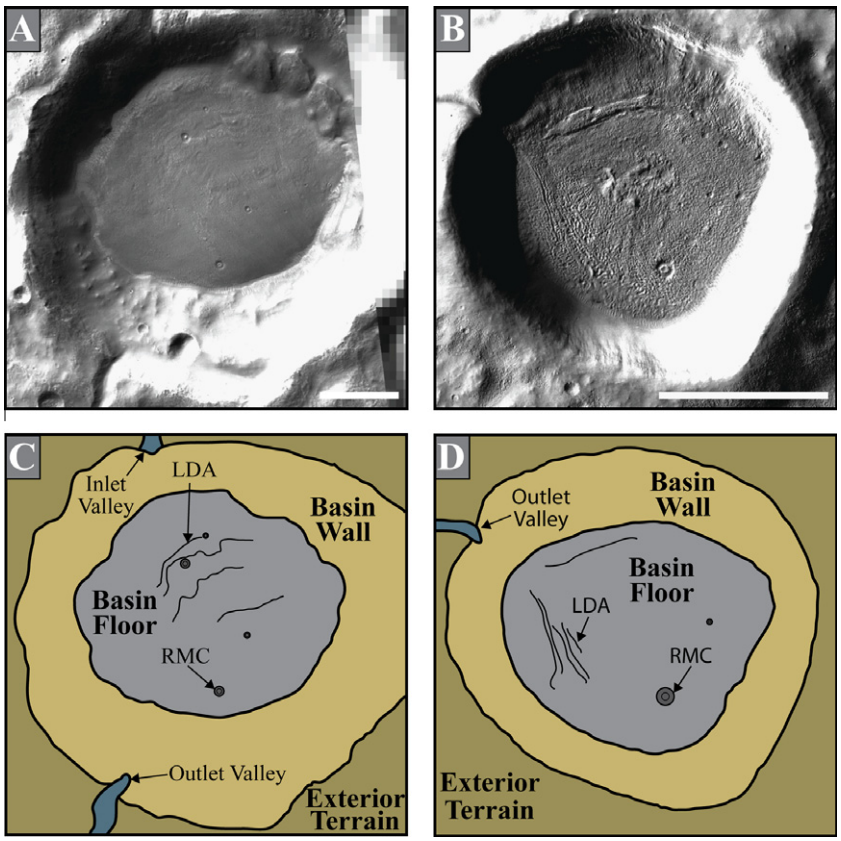


Fig. 5. Examples of glacially resurfaced open-basin lakes. Scale bars are 2 km. (A) Glacially resurfaced open-basin lake at 38.91°S, 102.96°W (Fassett and Head, 2008a). CTX image P15_006798_1405_XI_39S103 W overlain on THEMIS visible mosaic. (B) Glacially resurfaced open-basin lake at 36.24°S, 124.42°W (Fassett and Head, 2008a). CTX image B02_010596_1432_XN_36S124 W. Note that this is an unusual open-basin lake that lacks obvious input valleys. (C and D) Geologic sketch maps of the glacially resurfaced open-basin lakes in parts (A and B) respectively. Ring-mold craters (RMCs) and lobate debris aprons (LDAs) are indicated.

position controlled by the relative proportions of Fe and Mg (Bishop et al., 2002).

The identification of kaolinite is based on a strong hydroxyl overtone at 1.4 μm and a doublet absorption at 2.16 and 2.20 μm, as well as a weak absorption at 1.9 μm (Fig. 10). As with smectite, the 1.4 and 1.9 μm absorptions are due to an overtone and combination tone of fundamental absorptions of bound OH and H_2O respectively (Clark et al., 1990), while the 2.16 and 2.20 μm doublet is caused by combination tones of fundamental absorptions of Al–OH in the kaolinite structure (Clark et al., 1990).

Hydrated silica is identified based on diagnostic absorptions at 1.4 and 1.9 μm and a broad absorption centered at ~2.2 μm (Fig. 10). The bands at 1.4 and 1.9 μm are again related to fundamental absorptions from bound OH and H_2O, while the broad 2.2 μm absorption is caused by combination tones of fundamental absorptions from the Si–OH bond (Stolper, 1982).

Carbonate is identified based on paired absorptions at 2.31 and 2.51 μm, as well as an absorption at 1.9 μm (Fig. 10). The paired absorptions at 2.31 and 2.51 μm are caused by overtones of fundamental carbonate absorptions, while the 1.9 μm band is caused by a combination tone of fundamental absorptions of structurally bound H_2O (Hunt and Salisbury, 1971; Gaffey, 1987).

4. Implications of the global distribution of open-basin lake characteristics

Several interesting observations arise from the global distribution of features and processes implied by the classification presented here (Figs. 7 and 8). First, it is clear that there is a distinctly higher density of open-basin lakes with exposed sedimentary deposits in the Nili Fossae region (Fig. 7, outlined area). Indeed, all of the open-basin lakes previously identified in Nili Fossae contain some form of exposed deposit of possible lacustrine origin (Fig. 7; compare to Fig. 1 in Fassett and Head (2008a)). Two scenarios that might explain this observation are: (1) this area has been exhumed, leading to an exposure of the original sedimentary deposits within the open-basin lakes and/or (2) this area had a higher sediment load during open-basin lake lacustrine activity or was active for a longer period of time, thus creating thicker sedimentary deposits, which have been preferentially preserved and remain exposed. Although both of these mechanisms are possible, the exposed open-basin lake deposits in Nili Fossae are commonly observed below a partially eroded resurfacing unit, which is in agreement with previous work that cites exhumation as a dominant geologic process in this area (e.g. Mangold et al., 2007; Mustard et al., 2007; Harvey and Griswold, 2010). The magnitude of exhumation in the Nili Fossae region is also likely to have been substantial, as enough material has been eroded, possibly due to aeolian activity (Mangold et al., 2007), to expose ancient Noachian crustal material (Mangold et al., 2007; Mustard et al., 2007). Therefore, we tentatively favor the hypothesis of exhumation as the most plausible explanation for the high density of exposed sedimentary deposits in the Nili Fossae region.

The distribution shown by the resurfacing classification also reveals some interesting trends (Fig. 8). While the large number of volcanically resurfaced open-basin lakes (96; ~42%) is not surprising due to the fact that at least ~30% of the martian surface has been resurfaced by Hesperian volcanic flows (Head et al., 2002), the locations of the volcanically resurfaced open-basin lakes are interesting. Although many of the volcanically resurfaced open-

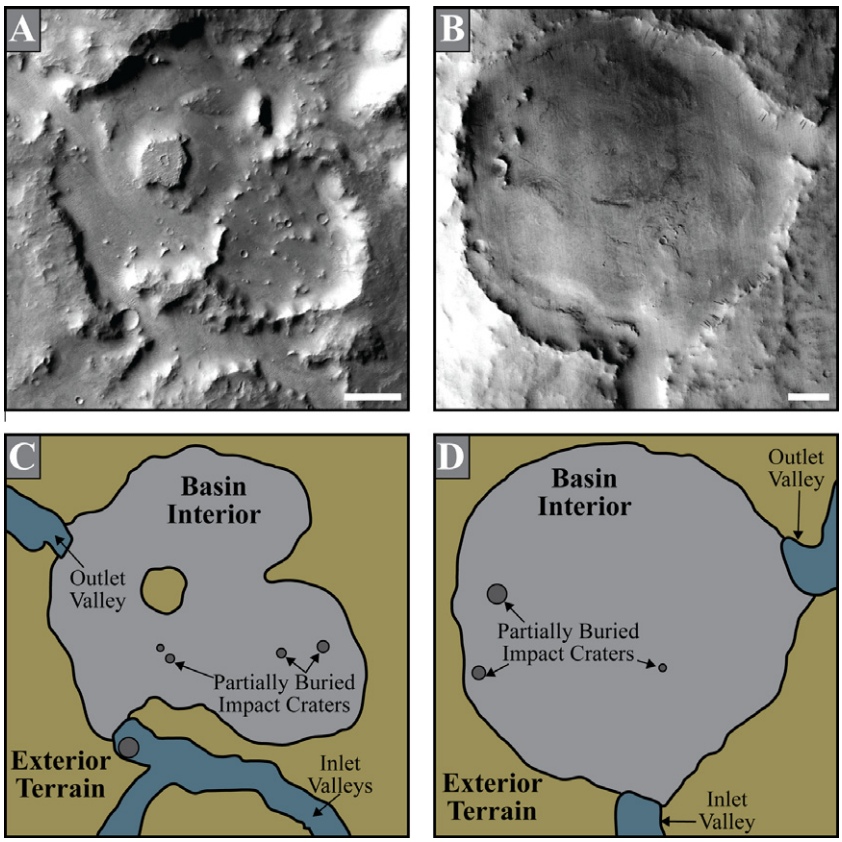


Fig. 6. Examples of resurfaced open-basin lakes with unknown source of resurfacing. Note the consistent texture both inside and outside the basin. Scale bars are 2 km. (A) Resurfaced open-basin lake at 0.61°N, 91.26°E (Fassett and Head, 2008a). CTX image P18_007885_1824_XN_02N269 W. (B) Resurfaced open-basin lake at 13.20°N, 19.52°E (Fassett and Head, 2008a). CTX image P13_005989_1932_XN_13N340 W. (C and D) Geologic sketch maps of the resurfaced open-basin lakes in parts (A and B) respectively. Partially buried impact craters are indicated.

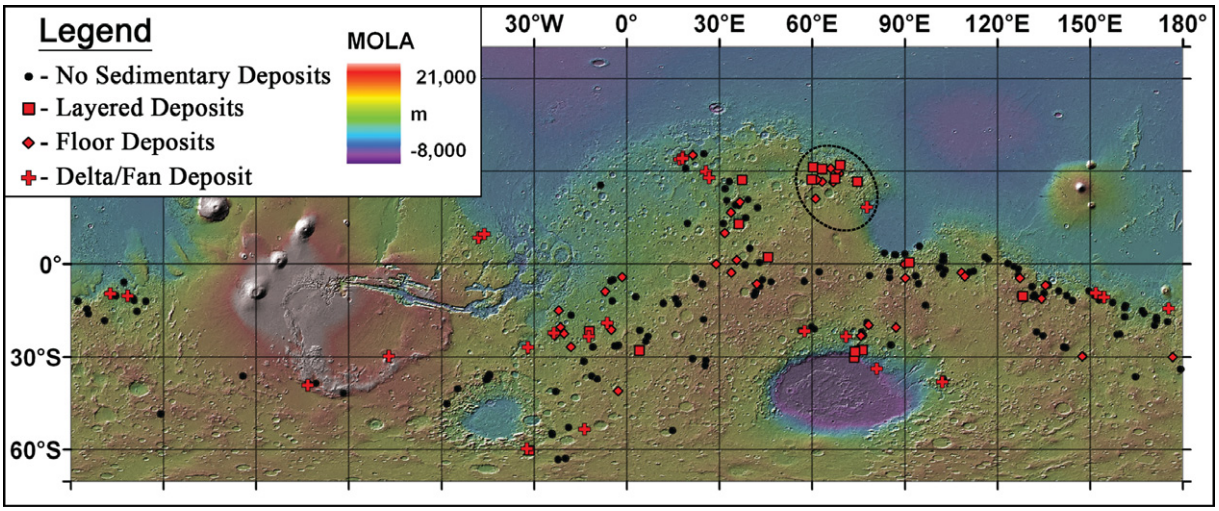


Fig. 7. Global distribution of the results of the exposed open-basin lake deposits classification. Area with high concentration of exposed open-basin lake deposits (Nili Fossae) is outlined in dotted black line. Background is MOLA topography overlain on MOLA hillshade (Smith et al., 2001).

basin lakes tend to cluster around clear volcanic sources such as Syrtis Major, Hesperia Planum and Apollinaris Mons, there also appears to be a clustering of volcanically resurfaced open-basin lakes in areas such as Arabia Terra and Margaritifer Terra (Fig. 8B, outlined areas), which are not linked to an established volcanic source. This lack of an obvious nearby volcanic vent requires more distributed volcanism not easy to trace to a specific edifice, such as large feeder dikes, which have been observed in Terra Tyrrhena and are thought to have emplaced regional expanses of Hesperian ridged plains (Head et al., 2006). Such dikes may also be responsible for floor-fractured craters, which may indicate past regional volcanic activity that modified pre-existing impact craters (Schultz, 1978; Schultz and Glicken, 1979). Indeed, we find that floor-fractured craters exist to the northwest of the cluster of

Table 1
Results of the exposed open-basin lake deposit classification and the open-basin lake resurfacing classification. Table lists: location of each open-basin lake; the type of imagery used for basin analysis; the type of identified exposed sedimentary deposit, where D = delta deposit, L = layered deposit, EFM = exposed floor material and N/A = no exposed deposit identified; the type of identified resurfacing, where V = volcanically resurfaced, G = glacially resurfaced and U = unknown source of resurfacing; the degree of confidence in the identification of the basin as a paleolake; and suitable references.

Basin number[a]	Open-basin lake location		Imagery used	Exposed deposit type	Resurfacing type	Confidence in basin identification	Reference(s)
	Lon. (E)	Lat. (N)					
1	116.86	1.46	THEMIS	N/A	U	Confident	Fassett and Head (2008a)
2	151.78	−9.29	CTX	D	V	Confident	Irwin et al. (2007)
3	166.75	−15.20	CTX	N/A	U	Probable	Fassett and Head (2008a)
4	−174.86	−14.63	CTX	N/A	V	Confident	Forsythe and Zimbelman (1995)
5	−161.57	−10.32	CTX	D	U	Confident	Cabrol and Grin (1999, 2001)
6	152.75	−11.53	HRSC	N/A	V	Confident	Irwin et al. (2007)
7	157.12	−12.44	CTX	N/A	V	Confident	Fassett and Head (2008a)
8	42.19	18.25	HRSC	N/A	U	Confident	Fassett and Head (2008a)
9	59.68	27.47	HRSC	L	U	Confident	Fassett and Head (2008a)
10	60.94	21.10	HRSC	EFM	V	Confident	Fassett and Head (2008a)
11	63.02	26.57	HRSC	EFM	U	Confident	Fassett and Head (2008a)
12	100.97	−1.31	CTX	N/A	V	Confident	Fassett and Head (2008a)
13	102.02	0.92	CTX	N/A	V	Probable	Fassett and Head (2008a)
14	102.36	2.42	CTX	N/A	V	Probable	Fassett and Head (2008a)
15	−169.20	−18.30	THEMIS	N/A	V	Confident	Cabrol and Grin (1999, 2001)
16	−12.32	−21.67	THEMIS	L	V	Confident	Goldspiel and Squyres (1991)
17	−8.59	25.57	HRSC	N/A	V	Confident	Fassett and Head (2008a)
18	−7.21	−8.84	HRSC	EFM	V	Confident	Fassett and Head (2008a)
19	2.77	−10.63	HRSC	N/A	V	Confident	Fassett and Head (2008a)
20	60.11	31.33	CTX	L	U	Confident	Fassett and Head (2008a)
21	63.03	30.85	HRSC	L	U	Confident	Fassett and Head (2008a)
22	78.43	−3.73	HRSC	N/A	V	Possible	Fassett and Head (2008a)
23	84.96	−2.42	HRSC	N/A	V	Confident	Fassett and Head (2008a)
24	84.96	−4.46	HRSC	N/A	V	Confident	Fassett and Head (2008a)
25	89.71	−0.09	CTX	EFM	V	Confident	Cabrol and Grin (1999, 2001)
26	90.01	−4.57	CTX	EFM	V	Confident	Cabrol and Grin (1999, 2001)
27	108.19	−2.84	CTX	EFM	V	Confident	Cabrol and Grin (1999, 2001)
28	109.22	−4.00	CTX	EFM	U	Confident	Fassett and Head (2008a)
29	110.86	−2.71	CTX	N/A	V	Confident	Cabrol and Grin (1999, 2001)
30	144.06	−11.74	CTX	N/A	V	Confident	Cabrol and Grin (1999, 2001)
31	154.52	−10.77	THEMIS	D	V	Confident	Fassett and Head (2008a)
32	−23.53	−23.12	CTX	EFM	V	Confident	Fassett and Head (2008a)
33	−5.32	−5.51	CTX	N/A	V	Confident	Cabrol and Grin (1999, 2001)
34	102.25	−3.42	HRSC	N/A	V	Probable	Fassett and Head (2008a)
35	3.88	−27.90	HRSC	L	V	Confident	Fassett and Head (2008a)
36	31.05	13.18	CTX	N/A	V	Probable	Irwin et al. (2005)
37	31.58	10.02	CTX	EFM	V	Confident	Irwin et al. (2005)
38	35.46	1.30	CTX	EFM	U	Confident	Cabrol and Grin (1999, 2001)
39	41.98	−6.37	CTX	EFM	V	Confident	Cabrol and Grin (1999, 2001)
40	25.57	29.91	CTX	D	G	Confident	McGill (2002)
41	31.72	24.39	THEMIS	N/A	V	Confident	Fassett and Head (2008a)
42	36.40	20.04	HRSC	EFM	U	Confident	Fassett and Head (2008a)
43	35.08	18.99	CTX	N/A	U	Confident	Fassett and Head (2008a)
44	33.57	16.72	CTX	EFM	V	Possible	Fassett and Head (2008a)
45	77.70	18.38	CTX	D	V	Confident	Fassett and Head (2005)
46	127.20	−10.32	THEMIS	N/A	U	Confident	Fassett and Head (2008a)
47	128.01	−10.37	CTX	L	V	Confident	Fassett and Head (2008a)
48	131.05	−7.35	THEMIS	N/A	U	Confident	Fassett and Head (2008a)
49	127.05	−4.58	CTX	EFM	V	Confident	Irwin et al. (2007)
50	175.39	−14.40	CTX	D	V	Confident	Grin and Cabrol (1997); Cabrol and Grin (1999, 2001)
51	176.56	−30.12	THEMIS	EFM	V	Confident	Irwin et al. (2002)

52	−48.00	8.54	CTX	D	U	Confident	Harrison and Grimm (2005), Hauber et al. (2009)
53	17.06	33.80	CTX	D	G	Confident	Cabrol and Grin (1999, 2001)
54	18.19	34.61	CTX	D	V	Confident	Cabrol and Grin (1999, 2001)
55	18.04	34.17	CTX	D	V	Confident	Cabrol and Grin (1999, 2001)
56	−155.77	−12.07	CTX	N/A	V	Confident	Fassett and Head (2008a)
57	−158.65	−15.45	THEMIS	N/A	U	Confident	Fassett and Head (2008a)
58	26.65	27.92	CTX	D	U	Confident	McGill (2002)
59	59.62	−20.31	CTX	N/A	U	Confident	Lahtela et al. (2005)
60	60.63	−20.99	CTX	N/A	U	Confident	Lahtela et al. (2005)
61	−102.96	−38.91	CTX	N/A	G	Confident	Fassett and Head (2008a)
62	−91.82	−41.86	CTX	N/A	U	Confident	Fassett and Head (2008a)
63	125.18	−0.81	HRSC	N/A	U	Confident	Irwin et al. (2007)
64	126.62	−1.60	CTX	N/A	V	Confident	Irwin et al. (2007)
65	−12.27	−23.44	CTX	D	V	Confident	Irwin et al. (2007)
66	−6.32	−19.36	CTX	EFM	V	Confident	Fassett and Head (2008a)
67	4.05	−21.49	CTX	N/A	V	Confident	Fassett and Head (2008a)
68	151.35	−10.28	HRSC	N/A	V	Possible	Irwin et al. (2007)
69	61.43	27.33	CTX	EFM	U	Confident	Fassett and Head (2008a)
70	135.38	−6.91	CTX	EFM	V	Confident	Fassett and Head (2008a)
71	136.47	−6.72	CTX	N/A	U	Probable	Fassett and Head (2008a)
72	134.92	−9.45	HRSC	N/A	V	Confident	Cabrol and Grin (1999, 2001)
73	134.07	−11.22	CTX	EFM	V	Confident	Fassett and Head (2008a)
74	−167.18	−9.54	CTX	D	U	Confident	Fassett and Head (2008a)
75	−165.61	−10.22	THEMIS	N/A	V	Confident	Cabrol and Grin (1999, 2001)
76	−162.87	−5.88	HRSC	N/A	U	Confident	Fassett and Head (2008a)
77	134.49	−23.21	HRSC	N/A	V	Confident	Fassett and Head (2008a)
78	123.25	0.07	CTX	N/A	U	Confident	Fassett and Head (2008a)
79	161.08	−14.00	HRSC	N/A	V	Confident	Fassett and Head (2008a)
80	159.46	−22.78	HRSC	N/A	V	Confident	Fassett and Head (2008a)
81	−1.73	−4.23	CTX	EFM	U	Confident	Fassett and Head (2008a)
82	62.17	−2.56	HRSC	N/A	V	Probable	Fassett and Head (2008a)
83	−18.29	−26.87	CTX	EFM	V	Confident	Fassett and Head (2008a)
84	87.00	−20.47	CTX	EFM	V	Confident	Fassett and Head (2008a)
85	96.52	−13.32	CTX	N/A	V	Confident	Fassett and Head (2008a)
86	56.22	−21.58	HRSC	N/A	U	Probable	Fassett and Head (2008a)
87	94.29	−6.29	CTX	N/A	U	Confident	Fassett and Head (2008a)
88	149.44	−8.68	CTX	N/A	V	Confident	Fassett and Head (2008a)
89	167.28	−16.01	HRSC	N/A	U	Confident	Cabrol and Grin (1999, 2001)
90	170.73	−19.98	THEMIS	N/A	V	Confident	Cabrol and Grin (1999, 2001)
91	171.33	−17.44	CTX	N/A	V	Confident	Cabrol and Grin (1999, 2001)
92	171.28	−18.21	HRSC	N/A	V	Confident	Fassett and Head (2008a)
93	16.44	−12.82	CTX	N/A	U	Confident	Cabrol and Grin (1999, 2001)
94	−13.68	−53.37	CTX	D	U	Confident	Fassett and Head (2008a)
95	−20.51	−22.45	CTX	EFM	V	Confident	Goldspiel and Squyres (1991)
96	147.24	−29.81	CTX	EFM	V	Confident	Fassett and Head (2008a)
97	147.58	−29.85	CTX	N/A	V	Confident	Fassett and Head (2008a)
98	78.13	−19.71	THEMIS	EFM	U	Confident	Fassett and Head (2008a)
99	77.62	−18.85	THEMIS	N/A	U	Confident	Fassett and Head (2008a)
100	71.01	−23.42	HRSC	D	U	Confident	Fassett and Head (2008a)
101	76.48	−23.09	HRSC	N/A	U	Confident	Fassett and Head (2008a)
102	75.60	−23.20	THEMIS	EFM	U	Confident	Fassett and Head (2008a)
103	75.91	−22.10	CTX	N/A	U	Confident	Fassett and Head (2008a)
104	78.17	−20.07	THEMIS	N/A	U	Confident	Fassett and Head (2008a)
105	−2.96	−41.09	THEMIS	EFM	V	Confident	Fassett and Head (2008a)
106	−22.31	−63.24	HRSC	N/A	U	Confident	Cabrol and Grin (1999, 2001)
107	−19.96	−62.90	HRSC	N/A	U	Confident	Cabrol and Grin (1999, 2001)
108	−24.35	−55.22	HRSC	N/A	U	Confident	Fassett and Head (2008a)
109	−24.39	−54.69	HRSC	N/A	U	Confident	Fassett and Head (2008a)
110	−18.98	−52.89	THEMIS	N/A	G	Confident	Fassett and Head (2008a)

(*continued on next page*)

Table 1 (*continued*)

Basin number[a]	Open-basin lake location		Imagery used	Exposed deposit type	Resurfacing type	Confidence in basin identification	Reference(s)
	Lon. (E)	Lat. (N)					
111	11.70	−12.71	CTX	N/A	U	Confident	Fassett and Head (2008a)
112	19.52	13.20	CTX	N/A	U	Confident	Fassett and Head (2008a)
113	−32.41	−59.68	CTX	D	U	Confident	Fassett and Head (2008a)
114	−9.79	−37.20	CTX	N/A	V	Probable	Fassett and Head (2008a)
115	74.42	26.76	CTX	L	U	Confident	Fassett and Head (2008a)
116	93.40	−3.84	CTX	N/A	V	Confident	Fassett and Head (2008a)
117	26.18	28.70	HRSC	N/A	G	Confident	Fassett and Head (2008a)
118	68.86	32.05	CTX	L	U	Confident	Fassett and Head (2008a)
119	68.58	29.33	CTX	EFM	U	Possible	Fassett and Head (2008a)
120	68.04	30.29	CTX	EFM	U	Confident	Fassett and Head (2008a)
121	−174.46	−16.10	CTX	N/A	V	Confident	Fassett and Head (2008a)
122	−31.70	−60.14	CTX	L	U	Confident	Fassett and Head (2008a)
123	−21.67	−20.42	CTX	EFM	V	Confident	Fassett and Head (2008a)
124	33.08	26.73	CTX	N/A	V	Confident	Fassett and Head (2008a)
125	36.11	18.88	HRSC	N/A	U	Confident	Fassett and Head (2008a)
126	57.62	−21.65	CTX	D	U	Confident	Fassett and Head (2008a)
127	115.88	2.22	HRSC	N/A	U	Confident	Fassett and Head (2008a)
128	80.91	−33.83	CTX	D	U	Confident	Cabrol and Grin (1999, 2001)
129	−4.87	−11.92	CTX	N/A	U	Confident	Fassett and Head (2008a)
130	−177.87	−11.99	CTX	N/A	U	Confident	Fassett and Head (2008a)
131	22.25	−5.02	CTX	N/A	U	Confident	Fassett and Head (2008a)
132	24.69	−17.98	HRSC	N/A	V	Confident	Fassett and Head (2008a)
133	18.96	30.95	CTX	N/A	G	Confident	Cabrol and Grin (1999, 2001)
134	161.26	−13.43	HRSC	N/A	V	Confident	Fassett and Head (2008a)
135	−103.31	−39.19	CTX	D	G	Confident	Mangold and Ansan (2006)
136	−100.69	−38.57	HRSC	N/A	V	Confident	Mangold and Ansan (2006)
137	85.38	−26.11	HRSC	N/A	U	Confident	Fassett and Head (2008a)
138	−6.34	−19.05	CTX	D	U	Confident	Irwin et al. (2005)
139	28.76	−0.03	CTX	EFM	V	Confident	Cabrol and Grin (1999, 2001)
140	38.08	−0.38	HRSC	N/A	V	Confident	Cabrol and Grin (1999, 2001)
141	−22.18	−15.03	CTX	EFM	U	Confident	Cabrol and Grin (1999, 2001)
142	−150.98	−48.54	THEMIS	N/A	U	Possible	Fassett and Head (2008a)
143	14.66	−53.90	THEMIS	N/A	U	Possible	Fassett and Head (2008a)
144	−11.46	−36.28	HRSC	N/A	U	Confident	Fassett and Head (2008a)
145	−14.02	−31.50	HRSC	N/A	U	Confident	Fassett and Head (2008a)
146	−23.58	−22.28	CTX	D	V	Confident	Fassett and Head (2008a)
147	−4.04	−26.48	HRSC	N/A	U	Confident	Fassett and Head (2008a)
148	−11.15	−26.81	THEMIS	N/A	V	Confident	Fassett and Head (2008a)
149	41.11	−10.00	HRSC	N/A	V	Confident	Fassett and Head (2008a)
150	41.54	−9.01	CTX	N/A	U	Confident	Fassett and Head (2008a)
151	66.65	26.43	CTX	EFM	U	Confident	Fassett and Head (2008a)
152	67.22	27.72	CTX	L	U	Confident	Fassett and Head (2008a)
153	86.64	3.06	CTX	N/A	U	Confident	Fassett and Head (2008a)
154	174.88	−18.66	CTX	N/A	V	Confident	Fassett and Head (2008a)
155	−159.65	−11.63	CTX	N/A	V	Confident	Fassett and Head (2008a)
156	−159.75	−11.24	CTX	N/A	V	Confident	Fassett and Head (2008a)
157	−54.89	−40.41	HRSC	N/A	U	Confident	Fassett and Head (2008a)
158	−58.38	−45.16	CTX	N/A	U	Confident	Fassett and Head (2008a)
159	−45.66	−37.48	CTX	N/A	G	Confident	Fassett and Head (2008a)
160	−45.17	−36.73	CTX	N/A	G	Confident	Fassett and Head (2008a)
161	−44.62	−36.24	CTX	N/A	G	Confident	Fassett and Head (2008a)
162	−44.63	−35.70	CTX	N/A	G	Confident	Fassett and Head (2008a)
163	−45.29	−37.44	CTX	N/A	G	Confident	Fassett and Head (2008a)
164	15.68	−11.16	CTX	N/A	U	Confident	Fassett and Head (2008a)

165	39.08	20.78	CTX	N/A	U	Confident	Fassett and Head (2008a)
166	24.35	−6.44	CTX	N/A	U	Confident	Fassett and Head (2008a)
167	102.77	−1.93	THEMIS	N/A	V	Confident	Fassett and Head (2008a)
168	−46.18	9.63	HRSC	D	U	Confident	Fassett and Head (2008a)
169	139.72	−8.64	CTX	N/A	U	Confident	Irwin et al. (2007)
170	142.36	−10.26	CTX	N/A	V	Confident	Irwin et al. (2007)
171	−5.25	−21.26	THEMIS	EFM	V	Confident	Irwin et al. (2007)
172	131.84	−10.05	THEMIS	N/A	U	Confident	Fassett and Head (2008a)
173	131.44	−10.54	CTX	N/A	U	Confident	Fassett and Head (2008a)
174	132.47	−10.73	THEMIS	N/A	U	Confident	Fassett and Head (2008a)
175	132.51	−22.03	THEMIS	N/A	U	Confident	Fassett and Head (2008a)
176	142.05	−27.06	THEMIS	N/A	U	Confident	Fassett and Head (2008a)
177	141.34	−26.86	HRSC	N/A	U	Confident	Fassett and Head (2008a)
178	37.10	27.33	CTX	L	U	Confident	Fassett and Head (2008a)
179	32.30	20.57	CTX	N/A	U	Confident	Fassett and Head (2008a)
180	179.15	−34.02	CTX	N/A	V	Confident	Fassett and Head (2008a)
181	46.56	−5.74	HRSC	N/A	U	Confident	Fassett and Head (2008a)
182	57.31	−5.59	CTX	N/A	U	Confident	Fassett and Head (2008a)
183	43.76	−5.13	HRSC	N/A	V	Confident	Fassett and Head (2008a)
184	33.60	−2.76	HRSC	EFM	V	Confident	Fassett and Head (2008a)
185	45.51	2.16	HRSC	L	U	Confident	Fassett and Head (2008a)
186	42.91	0.52	HRSC	N/A	U	Confident	Fassett and Head (2008a)
187	42.99	−7.16	CTX	N/A	V	Confident	Fassett and Head (2008a)
188	36.12	13.12	THEMIS	L	U	Confident	Fassett and Head (2008a)
189	38.37	14.88	THEMIS	N/A	U	Confident	Fassett and Head (2008a)
190	−18.15	−16.51	CTX	N/A	V	Confident	Fassett and Head (2008a)
191	6.05	−24.92	THEMIS	N/A	U	Confident	Fassett and Head (2008a)
192	6.71	−23.45	CTX	N/A	V	Confident	Fassett and Head (2008a)
193	21.96	−4.49	CTX	N/A	U	Confident	Fassett and Head (2008a)
194	86.92	2.96	CTX	N/A	U	Confident	Fassett and Head (2008a)
195	83.20	3.51	CTX	N/A	U	Confident	Fassett and Head (2008a)
196	91.26	0.61	CTX	L	U	Confident	Fassett and Head (2008a)
197	89.84	3.25	CTX	N/A	V	Confident	Fassett and Head (2008a)
198	94.61	5.81	CTX	N/A	U	Confident	Fassett and Head (2008a)
199	158.54	−22.89	HRSC	N/A	V	Confident	Fassett and Head (2008a)
200	102.49	1.08	HRSC	N/A	U	Confident	Fassett and Head (2008a)
201	112.13	−2.39	HRSC	N/A	V	Possible	Fassett and Head (2008a)
202	39.74	5.16	HRSC	N/A	V	Confident	Fassett and Head (2008a)
203	126.63	−3.77	CTX	N/A	U	Confident	Fassett and Head (2008a)
204	−23.19	−41.24	CTX	N/A	G	Confident	Fassett and Head (2008a)
205	25.51	−31.11	CTX	N/A	V	Confident	Fassett and Head (2008a)
206	21.21	−30.59	HRSC	N/A	U	Confident	Fassett and Head (2008a)
207	25.25	−32.75	CTX	N/A	G	Confident	Fassett and Head (2008a)
208	160.98	−17.11	HRSC	N/A	V	Confident	Fassett and Head (2008a)
209	101.58	−0.90	CTX	N/A	U	Confident	Fassett and Head (2008a)
210	−4.52	−5.08	CTX	N/A	U	Confident	Fassett and Head (2008a)
211	73.68	−28.25	CTX	L	U	Confident	Wilson et al. (2007)
212	76.11	−27.72	CTX	L	U	Confident	
213	65.87	30.87	CTX	EFM	U	Confident	
214	−32.07	−26.99	CTX	D	U	Confident	Buhler et al. (2011)
215	73.38	−30.12	CTX	L	U	Confident	
216	21.68	35.06	CTX	N/A	G	Confident	
217	24.84	35.58	CTX	N/A	G	Confident	
218	−124.42	−36.24	CTX	N/A	G	Confident	
219	102.30	−37.16	HRSC	N/A	G	Confident	
220	−103.26	−38.76	CTX	N/A	G	Confident	
221	164.51	−36.52	CTX	N/A	U	Confident	
222	−26.38	−3.22	CTX	N/A	V	Confident	Irwin et al. (2007)
223	−26.29	−2.86	CTX	N/A	V	Confident	Irwin et al. (2007)

(*continued on next page*)

Table 1 (*continued*)

Basin number[a]	Open-basin lake location		Imagery used	Exposed deposit type	Resurfacing type	Confidence in basin identification	Reference(s)
	Lon. (E)	Lat. (N)					
224	102.15	−38.17	HRSC	D	G	Confident	
225	21.21	35.21	CTX	EFM	G	Confident	
226	−77.09	−29.76	CTX	D	U	Confident	

[a] Basin numbers are after Fassett and Head (2008a), up to basin number 210.

volcanically resurfaced open-basin lakes in Margaritifer Terra (Schultz, 1978; Schultz and Glicken, 1979; Sato et al., 2010), and to the northeast of the cluster of volcanically resurfaced open-basin lakes in Arabia Terra (Fig. 8B) (Schultz, 1978; Schultz and Glicken, 1979). The presence of these features is thought to be related to regional volcanic activity (Schultz, 1978; Schultz and Glicken, 1979), and may help explain the areas of anomalous volcanically resurfaced open-basin lakes through the emplacement of widely distributed flood volcanism. However, more work is required to determine the exact source of these units.

All of the open-basin lakes that have been resurfaced by glacial activity are confined to mid- to high-latitudes (poleward of ±∼30°) (Fig. 8C). This distribution is consistent with the latitude dependence of the majority of these ice-related deposits (e.g. Head et al., 2003, 2010; Milliken et al., 2003; Levy et al., 2010), with the exception of a few unusual outliers at lower latitudes (Shean, 2010). Furthermore, while the amount of glacially resurfaced open-basin lakes (21) is substantially smaller than the number resurfaced by volcanic activity (96), the open-basin lakes poleward of ±30° are much more affected by glacial resurfacing than volcanic resurfacing (Fig. 8). This observation reflects the latitude dependence of such ice-related deposits, and is consistent with the occurrence of episodes of Late Amazonian glaciation on the surface of Mars (e.g. Mustard et al., 2001; Head et al., 2003, 2008; Schon et al., 2009; Dickson et al., 2010, 2011), post-dating the predominantly Hesperian-aged volcanic plains (Scott and Tanaka, 1986; Greeley and Guest, 1987).

5. Compositional implications for lacustrine deposits

Approximately one-third of the investigated open-basin lakes contain exposed sedimentary deposits, which might be expected to contain aqueous alteration and/or evaporite minerals as is commonly observed on Earth (Jones and Bowser, 1978; Blatt et al., 1980; Leeder, 1999; Wetzel, 2001). In terrestrial lacustrine deposits, aqueous alteration minerals (e.g. phyllosilicates, zeolites, hydrated silica) can occur both as transported sediment and as authigenic material (Moore, 1961; Muller and Quakernaat, 1969; Singer et al., 1972; Jones and Bowser, 1978; Blatt et al., 1980; Jones, 1986; Hay et al., 1991; Hillier, 1993). On Earth, transported sediment is likely to be rich in phyllosilicates because these are typically the most common alteration minerals produced in chemical weathering, and thus will be abundant in an active watershed (Moore, 1961; Muller and Quakernaat, 1969; Singer et al., 1972; Jones and Bowser, 1978; Blatt et al., 1980; Nesbitt and Young, 1989). Aqueous alteration minerals such as phyllosilicates and zeolites are also common products of chemical alteration caused by lithification and diagenesis of lacustrine sediments (Jones and Bowser, 1978; Blatt et al., 1980; Jones, 1986; Hay et al., 1991; Hillier, 1993). Evaporites in terrestrial lacustrine systems are almost exclusively confined to being precipitated *in situ* (Brunskill, 1969; Eugster and Hardie, 1975, 1978; Jones and Bowser, 1978; Kelts and Hsu, 1978; Strong and Eadie, 1978; Wetzel, 2001), although carbonate minerals may be present as transported material, especially in carbonate rich terrain (e.g. Coakley and Rust, 1968; Jones and Bowser, 1978; Kelts and Hsu, 1978; Ingles and Anadon, 1991). The typical sequence of evaporite formation is carbonates when the lake is deep and has a higher influx of surface water, followed by sulfates and chlorides as the lake is filled in and becomes less dominated by surficial flows (Eugster and Hardie, 1975, 1978; Jones and Bowser, 1978; Kelts and Hsu, 1978; Wetzel, 2001). Terrestrial lacustrine sediments from large, open systems are typically rich in clay minerals (>∼15–25 wt.%), and contain variable amounts of carbonate minerals depending on the water chemistry of the lake system (e.g. Moore, 1961; Thomas, 1969; Muller and

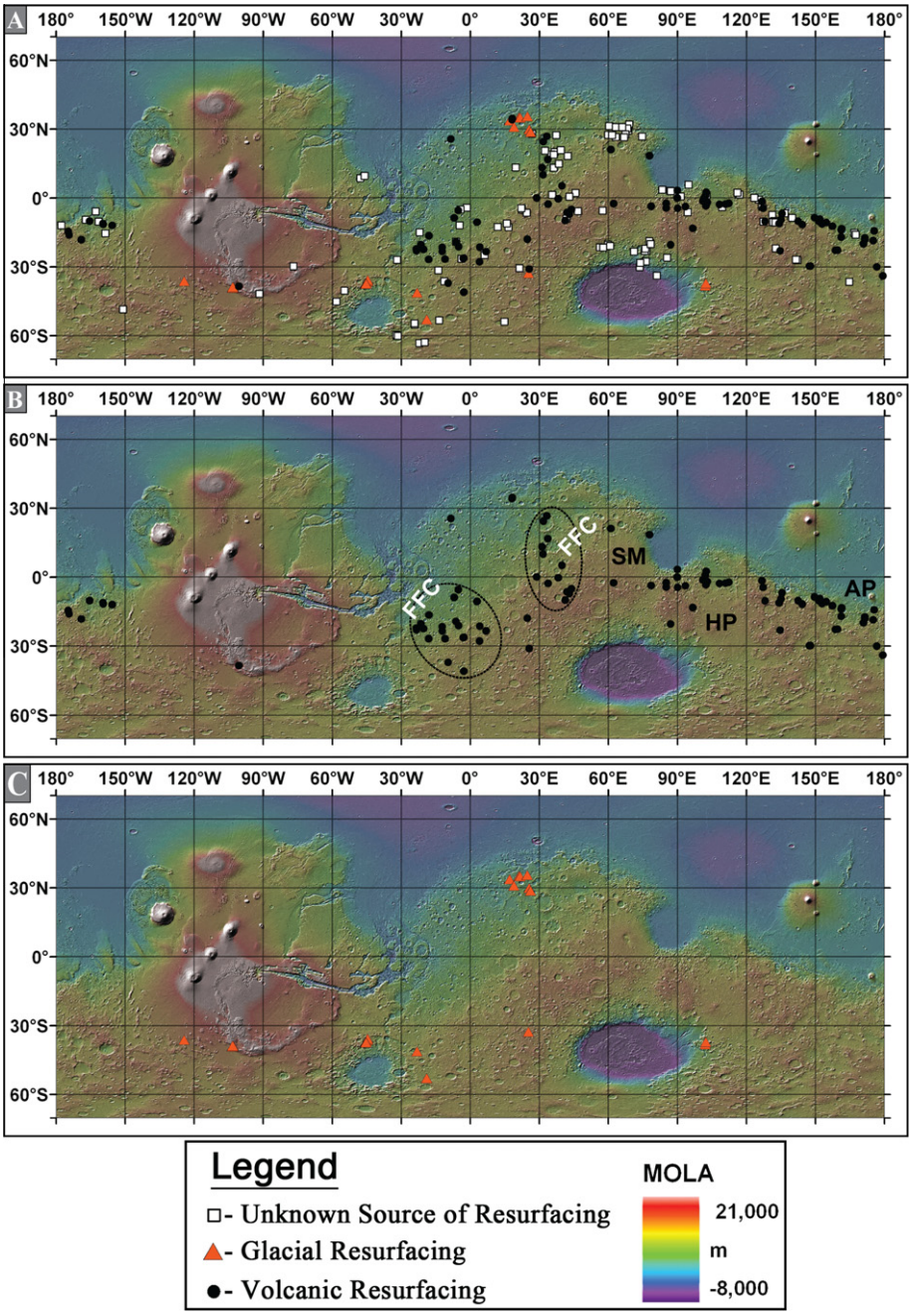


Fig. 8. Global distribution of the results of the open-basin lake resurfacing classification. Background is MOLA topography overlain on MOLA hillshade (Smith et al., 2001). Legend at bottom indicates resurfacing types. (A) Summary of all of the results for the open-basin lake resurfacing classification. (B) Global distribution of volcanically resurfaced open-basin lakes. Major volcanic sources are indicated in black text, SM = Syrtis Major, HP = Hesperia Planum and AP = Apollinaris Patera. Areas with high concentrations of volcanically resurfaced open-basin lakes but no obvious volcanic source (Arabia Terra and Margaritifer Terra) are outlined in dotted black lines. Areas of floor-fractured craters (FFCs) in Margaritifer Terra and Arabia Terra are indicated in white text. (C) Global distribution of glacially resurfaced open-basin lakes.

Quakernaat, 1969; Singer et al., 1972; Thomas et al., 1972, 1973; Jones and Bowser, 1978; Blatt et al., 1980; Hillier, 1993). As a direct example, sediments examined from the North American Great Lakes, which are comparable to the largest martian open-basin lakes in surface area and volume (Fassett and Head, 2008a), contain, on average, from ~15 to 70 wt.% clay minerals and ~0–40 wt.% carbonate minerals (e.g. Moore, 1961; Thomas, 1969; Thomas et al., 1972, 1973; Jones and Bowser, 1978).

Based solely on these terrestrial observations, one would expect that exposed sedimentary deposits observed in open-basin lakes on Mars should have a composition that includes appreciable amounts of aqueous alteration minerals and carbonates (e.g. Moore, 1961; Thomas, 1969; Muller and Quakernaat, 1969; Singer et al., 1972; Thomas et al., 1972, 1973; Jones and Bowser, 1978; Blatt et al., 1980; Hillier, 1993). These deposits would also be expected to contain chlorides and sulfates if the lacustrine systems were active for long enough to become heavily infilled and more playa-like (Eugster and Hardie, 1975, 1978; Jones and Bowser, 1978; Kelts and Hsu, 1978; Wetzel, 2001). The exact composition of these martian evaporite deposits would be expected to differ from those observed on Earth, however, due to different source water compositions (Tosca and McLennan, 2006). While recent

Table 2
Exposed open-basin lake deposits with targeted CRISM coverage. Table lists the basin number, from Table 1, the basin location, the exposed deposit type contained within the open-basin lake, and the CRISM ID number for the observation over the identified exposed sedimentary deposit.

Basin number	Open-basin lake location		Exposed open-basin lake deposit type	CRISM ID number
	Lon. (E)	Lat. (N)		
10	60.94	21.10	Exposed floor materials	FRT00007CBB
11	63.02	26.57	Exposed floor materials	FRT00008474
16	−12.32	−21.67	Layered deposits	HRS0000AF77
21	63.03	30.85	Layered deposits	FRT00004509
25	89.71	−0.09	Exposed floor materials	FRT000171B3
27	108.19	−2.84	Exposed floor materials	HRL00013417
28	109.22	−4.00	Exposed floor materials	FRT0001BBA7
37	31.58	10.02	Exposed floor materials	FRT0000BC8E
39	41.98	−6.37	Exposed floor materials	HRL0000AA9F
40	25.57	29.91	Delta	FRT0001716C
44	33.57	16.72	Exposed floor materials	FRT0000A859
45	77.70	18.38	Delta	HRL000040FF
51	176.56	−30.12	Exposed floor materials	FRT00008C90
52	−48.00	8.54	Delta	FRT000098B5
53	17.06	33.80	Delta	HRS0000CD2D
58	26.65	27.92	Delta	FRT0001648B
65	−12.27	−23.44	Delta	HRL00019A9F
83	−18.29	−26.87	Exposed floor materials	FRT0000C9DB
94	−13.68	−53.37	Delta	FRT00008999
95	−20.51	−22.45	Exposed Floor Materials	HRS0000AC56
105	−2.96	−41.09	Exposed Floor Materials	FRT00007536, FRT000099C7
115	74.42	26.76	Layered deposits	HRL0000AB0A
123	−21.67	−20.42	Exposed floor materials	FRT00017BA7
135	−103.31	−39.19	Delta	FRT0000944A
139	28.76	−0.03	Exposed floor materials	FRT00010832
151	66.65	26.43	Exposed floor materials	FRT0000ADBC
152	67.22	27.72	Layered deposits	FRT0000A977
171	−5.25	−21.26	Exposed floor materials	FRT0000A4C4
188	36.12	13.12	Layered deposits	FRT0000A41C
211	73.68	−28.25	Layered deposits	FRT0000622B
212	76.11	−27.72	Layered deposits	FRT000091B1
213	65.87	30.87	Exposed floor materials	FRT000136F2
214	−32.07	−26.99	Delta	FRT000163EE
215	73.38	−30.12	Layered deposits	HRL000A153

Table 3
Exposed open-basin lake deposits containing identified aqueous alteration and evaporite minerals. Table lists open-basin lake number, from Table 1, the basin location, type of exposed deposit, where D = delta deposit, L = layered deposit and EFM = exposed floor material, identified mineral(s), CRISM ID number and reference(s) for identified aqueous alteration and evaporite minerals.

Basin number	Open-basin lake location		Exposed deposit type	Detected mineral(s)	CRISM ID	Reference(s)
	Lon. (E)	Lat. (N)				
10	60.94	21.10	EFM	Hydrated silica, Fe/Mg–smectite	FRT00007CBB	Bandfield (2006), Ehlmann et al. (2009)
45	77.70	18.38	D	Fe/Mg–smectite, carbonate	HRL000040FF	Ehlmann et al. (2008a,b)
51	176.56	−30.12	EFM	Fe/Mg–smectite	FRT00008C90	Wray et al. (2009)
53	17.06	33.80	D	Fe/Mg–smectite	HRS0000CD2D	Dehouck et al. (2010)
83	−18.29	−26.87	EFM	Kaolinite	FRT0000C9DB	Goudge et al. (2011)
115	74.42	26.76	L	Fe/Mg–smectite	HRL0000AB0A	
123	−21.67	−20.42	EFM	Fe/Mg–smectite	FRT00017BA7	Goudge et al. (2011)
135	−103.31	−39.19	D	Fe/Mg–smectite	FRT0000944A	Hughes et al. (2011)
211	73.68	−28.25	L	Fe/Mg–smectite	FRT0000622B	Ansan et al. (2011)
212	76.11	−27.72	L	Fe/Mg–smectite	FRT00091B1	

Table 4
Location of obtained spectra from Fig. 10. Table lists open-basin lake number, from Table 1, identified mineral, CRISM ID number, center locations for numerator and denominator spectra, and number of pixels used for both numerator and denominator. All spectra displayed in Fig. 10 are a ratio of the numerator and denominator spectra at the given locations.

Basin number	Identified mineral	CRISM ID	Numerator center		Denominator center		Number of pixels
			X	*Y*	*X*	*Y*	
10	Hydrated silica	FRT00007CBB	204	377	204	326	7 × 7
45	Carbonate	HRL000040FF	280	132	280	377	7 × 7
51	Fe/Mg–smectite	FRT00008C90	315	229	315	177	9 × 9
83	Kaolinite	FRT0000C9DB	310	224	310	390	31 × 31
115	Fe/Mg–smectite	HRL0000AB0A	132	198	132	25	7 × 7

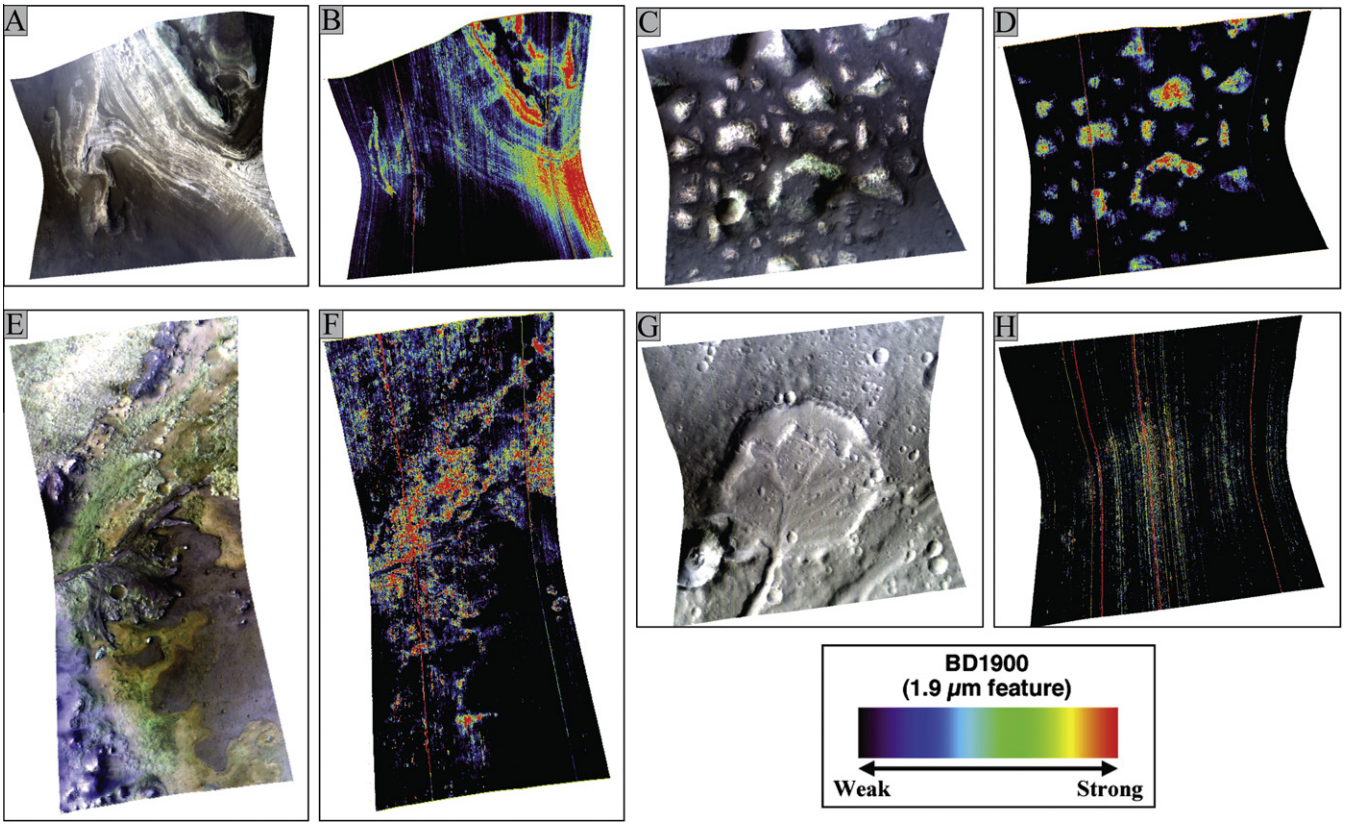


Fig. 9. CRISM observations showing the association between exposed open-basin lake deposits and aqueous alteration minerals, indicated by the BD1900 parameter (B, D, F, H), which relates to the strength of the 1.9 μm absorption (Pelkey et al., 2007). Stretch for all CRISM false color RGB images (A, C, E, G) is R = 2.3906 μm, G = 1.8904 μm and B = 1.1521 μm. Scale for the parameter values is indicated in the lower right. (A) FRT0000622B showing layering in an open-basin lake in Terby Crater, 28.25°S, 73.68°E (Wilson et al., 2007). (B) BD1900 parameter for FRT0000622B indicating the presence of hydrated minerals in the deposit shown in part (A) (Ansan et al., 2011). (C) FRT00008C90 showing exposed floor material in an open-basin lake at 30.12°S, 176.56°E (Irwin et al., 2002). (D) BD1900 parameter for FRT00008C90 indicating the presence of hydrated minerals in the deposit shown in part (C) (Wray et al., 2009). (E) HRL000040FF showing a delta deposit in an open-basin lake in Jezero Crater, 18.38°N, 77.70°E (Fassett and Head, 2005; Ehlmann et al., 2008a). (F) BD1900 parameter for HRL000040FF indicating the presence of hydrated minerals in the deposit shown in part (E) (Ehlmann et al., 2008a). (G) FRT000098B5 showing a delta deposit in an open-basin lake at 8.54°N, 48.00°W (Harrison and Grimm, 2005; Hauber et al., 2009). Note the lack of spectral diversity across the deposit. (H) BD1900 parameter for FRT000098B5 indicating the lack of hydrated minerals in the deposit shown in part (G).

work has shown that some martian paleo-lacustrine deposits do contain aqueous alteration and evaporite minerals (e.g. Ehlmann et al., 2008a, 2008b, 2009; Mustard et al., 2008; Dehouck et al., 2010; Ansan et al., 2011), our results indicate these occurrences are limited (~29% of exposed deposits of possible lacustrine origin with CRISM coverage have mineralogies that reflect aqueous alteration). Additionally, the only site with observed evaporite deposits (carbonate) is in Jezero crater, where it has been shown that the observed carbonate and Fe/Mg-smectite are present as transported material (Ehlmann et al., 2008a, 2008b).

These results indicate a divergence from the mineral assemblages that would be predicted for martian lacustrine deposits based on the study of terrestrial lacustrine systems (e.g. Moore, 1961; Thomas, 1969; Muller and Quakernaat, 1969; Singer et al., 1972; Thomas et al., 1972, 1973; Jones and Bowser, 1978; Blatt et al., 1980; Hillier, 1993), and could be due to three possible scenarios: (1) thin resurfacing units, such as aeolian dust cover, obscuring aqueous alteration and evaporite minerals within the exposed deposits from spectral detection and characterization from orbit; (2) available and currently analyzed spectral data at adequate resolution over these exposed deposits are too few in number to discern true trends; or (3) many exposed open-basin lake deposits have a lithology that is primarily composed of unaltered material.

Clearly, the first two mechanisms cannot be ruled out and could be contributing to the observed disparity. For the first mechanism in particular, there does seem to be some correlation between exposed open-basin lake deposits containing identified aqueous alteration minerals and fairly dust free areas indicated by a global dust cover index (Ruff and Christensen, 2002) (Fig. 11); however, there are a large number of exposed open-basin lake deposits also in these dust free areas that show no evidence for the presence of aqueous alteration minerals. Furthermore, we have identified aqueous alteration minerals in exposed open-basin lake deposits where there is both substantial regional dust cover as well as in areas that are largely dust free (Ruff and Christensen, 2002) (Fig. 11). It is important to note that the scale of the Ruff and Christensen (2002) dust cover index is much larger than the scale of many of the exposed sedimentary deposits examined here, and so more local dust cover might be present at those sites deemed to be dust free; however, we hypothesize that it is doubtful that a spectrally obscuring dust unit is the only cause for the disparity identified here, although it may be a contributing factor. Additionally, 34 of the 79 (~43%) exposed open-basin lake deposits have CRISM targeted observation coverage at a resolution that is adequate for identifying aqueous alteration and evaporite minerals (Table 2) (e.g. Ehlmann et al., 2008a, 2008b, 2009; Mustard et al., 2008; Dehouck et al., 2010); however, only 10 (~29% of the deposits with adequate spectral coverage) exposed deposits have such identified minerals. Therefore, it is doubtful that insufficient CRISM coverage is the only cause of the observed disparity. The final hypothesis that most open-basin lake deposits have an unaltered

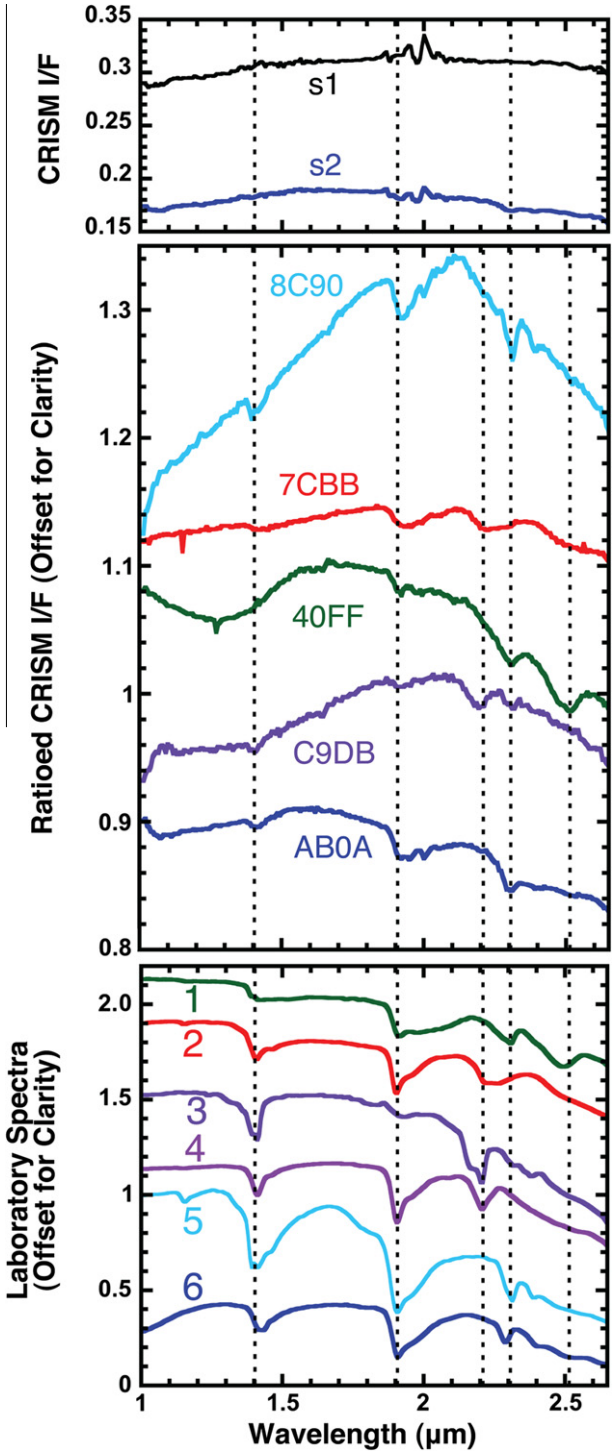


Fig. 10. Examples of the various types of aqueous alteration and evaporite mineral signatures observed in exposed open-basin lake deposits. See Table 4 for full CRISM image IDs and pixel locations. Top plot shows example numerator (s2) and denominator (s1) spectra used for rationing data from CRISM observation FRT0000AB0A. Numerator is characteristic of Fe/Mg-smectite within an exposed open-basin lake deposit while denominator is spectrally bland material. Dashed lines are located at 1.40, 1.91 and 2.30 μm. Middle plot shows ratioed spectra from five separate exposed open-basin lake deposits showing the range of aqueous alteration and evaporite minerals observed in such deposits. Spectra are offset for clarity. Partial CRISM observation IDs are given in the plot, see Table 4 for full observation information. Dashed lines are located at 1.40, 1.91, 2.20, 2.30 and 2.51 μm. Bottom plot shows laboratory spectra (Clark et al., 2007) of a suite of aqueous alteration and evaporite minerals, including: (1) Magnesite, Sample HS47.3B; (2) Hydrated Silica, Sample TM8896 (Hyalite); (3) Kaolinite, Sample KGa-2 (pxl); (4) Montmorillonite, Sample SWy-1; (5) Saponite, Sample SapCa-1; and (6) Nontronite, Sample NG-1.a. Dashed lines are located at 1.40, 1.91, 2.20, 2.30 and 2.51 μm.

lithology is a strong possibility; however, it is very difficult to assess, as the majority of the analyzed sedimentary deposits appear to have minimal spectral contrast with their surrounding terrain (Fig. 9G and H) and so it is difficult to accurately characterize their composition. While this effect could be due to either a local mantling or dust cover unit, or a similar composition between the sedimentary deposits and their surrounding terrain, it is difficult to definitively determine between these possibilities. It should also be noted that a small fraction of these deposits may have initially contained aqueous alteration and/or evaporite minerals that were subsequently removed due to a combination of poor deposit cementation and aeolian stripping.

Based on the results presented here, it is clear that the association between aqueous alteration minerals, evaporite minerals and lacustrine deposits is far more common on Earth (e.g. Moore, 1961; Thomas, 1969; Muller and Quakernaat, 1969; Singer et al., 1972; Thomas et al., 1972, 1973; Jones and Bowser, 1978; Blatt et al., 1980; Hillier, 1993) than it is on Mars. This observation suggests that *in situ* aqueous alteration and mineral precipitation was not as common in martian open-basin lakes as it is in terrestrial lakes, where mineral precipitation of carbonates and production of aqueous alteration minerals through diagenesis are common processes (Brunskill, 1969; Eugster and Hardie, 1975, 1978; Jones and Bowser, 1978; Kelts and Hsu, 1978; Strong and Eadie, 1978; Blatt et al., 1980; Jones, 1986; Hay et al., 1991; Hillier, 1993; Wetzel, 2001). If the aqueous alteration and evaporite minerals observed within the exposed sedimentary deposits were not formed *in situ*, then they must be transported in nature. The observed disparity is then reflective of the fact that aqueous alteration and evaporite minerals, while widespread, are only present in discrete locations across the martian surface (e.g. Gendrin et al., 2005; Poulet et al., 2005; Mustard et al., 2008; Osterloo et al., 2008, 2010; Ehlmann et al., 2011). Furthermore, the lack of observed evaporites, especially carbonates, in exposed open-basin lake deposits is consistent with the hypothesis that the input of water into open-basin lakes was brief and declined rapidly (Irwin et al., 2005), which would inhibit the buildup of dissolved ions, reducing the amount of *in situ* evaporite mineral precipitation. This conclusion is also in agreement with the suggestion of previous workers that the aqueous alteration minerals identified in individual open-basin lake deposits, and the carbonate observed in Jezero crater, are present as transported material (e.g. Ehlmann et al., 2008a,b; Dehouck et al., 2010; Ansan et al., 2011).

Further support for a transported origin for the aqueous alteration minerals is the distribution of identified minerals (Table 3). While we have identified kaolinite, hydrated silica, carbonate and Fe/Mg-smectite in exposed open-basin lake deposits, all of these minerals occur in only one deposit with the exception of Fe/Mg-smectite, which is found in nine of the ten exposed open-basin lake deposits. This distribution reflects the fact that Fe/Mg-smectite is the most common aqueous alteration mineral observed on the martian surface (e.g. Poulet et al., 2005; Mustard et al., 2008; Ehlmann et al., 2011), and suggests that the type of aqueous alteration minerals observed in these exposed open-basin lake deposits is dependent on the composition of the bedrock in the watershed. This correspondence between the mineralogy of lacustrine deposits and their watershed bedrock has been shown by previous workers for the Jezero delta deposit (Ehlmann et al., 2008a) and is commonly observed in lacustrine deposits on Earth (Moore, 1961; Muller and Quakernaat, 1969; Singer et al., 1972; Jones and Bowser, 1978; Blatt et al., 1980).

The observations presented here suggest that the observed aqueous alteration minerals within exposed open-basin lake deposits are transported in nature and that there was minimal *in situ* alteration or mineral precipitation occurring within these open-basin lakes. This is different than most lacustrine systems

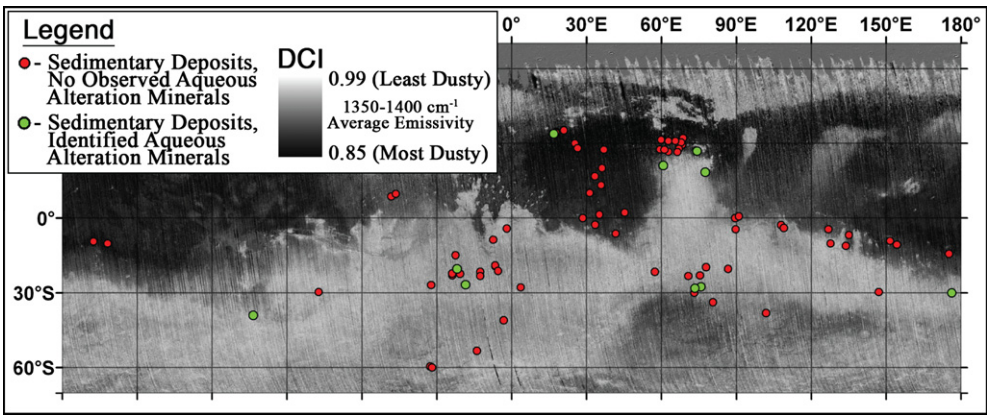


Fig. 11. Global distribution of exposed open-basin lake deposits compared with such deposits containing identified aqueous alteration minerals. Background is global dust cover index (DCI) (Ruff and Christensen, 2002) overlain on MOLA hillshade topography (Smith et al., 2001).

on Earth, where evaporites and aqueous alteration minerals are commonly produced *in situ* (Brunskill, 1969; Eugster and Hardie, 1975, 1978; Jones and Bowser, 1978; Kelts and Hsu, 1978; Strong and Eadie, 1978; Blatt et al., 1980; Jones, 1986; Hay et al., 1991; Hillier, 1993; Wetzel, 2001), which suggests differing conditions on Mars that were not conducive to *in situ* alteration and mineral precipitation. Two possible explanations for the lack of *in situ* alteration and mineral precipitation are: (1) a water chemistry that was not conducive to aqueous alteration of transported sediment and the precipitation of carbonates and/or (2) a geologically short-lived period of lacustrine activity within the open-basin lake systems. A geologically short period of lacustrine activity is also supported by the lack of chlorides and sulfates within these open-basin lake deposits, both of which have been identified on the surface of Mars (e.g. Gendrin et al., 2005; Osterloo et al., 2008, 2010; Bishop et al., 2009; Glotch et al., 2010). These evaporites are typically the last to develop within a lacustrine system, as the basin becomes less dominated by surficial flows and more playa-like as it is filled in with sediment (Eugster and Hardie, 1975, 1978; Jones and Bowser, 1978; Kelts and Hsu, 1978; Wetzel, 2001). The observation that chlorides and sulfates are not commonly observed in martian open-basin lakes suggests that the lacustrine systems were not active long enough to become infilled and begin precipitating these evaporites. The short-lived nature of many martian paleolake systems has been proposed previously based on observed morphologies (e.g. Howard et al., 2005; Irwin et al., 2005), and is supported by the findings of the work presented here; however, ruling out an unsuitable water chemistry is not possible with the data presented here.

6. Conclusions

Based on a detailed morphologic analysis of 226 open-basin lakes, approximately one-third (~35%) of these paleolakes contain exposed sedimentary deposits of possible lacustrine origin; however, all of the open-basin lakes are at least partially resurfaced, with volcanic resurfacing the most abundant resurfacing process identified. These findings support earlier, less detailed observations that post-lacustrine resurfacing has extensively modified the open-basin lake record on Mars (Fassett and Head, 2008a). Based on the geographic distribution of volcanically resurfaced open-basin lakes, many appear to have local volcanic sources (e.g. Syrtis Major, Hesperia Planum), while some do not and require more distributed sources of volcanic resurfacing such as dikes. This analysis also revealed a noticeable cluster of open-basin lakes with exposed sedimentary deposits in Nili Fossae. This suggests either exhumation or more significant lacustrine activity in this area, with an episode of exhumation being the preferred explanation. Furthermore, the distribution of glacially resurfaced open-basin lakes is consistent with both the latitude dependence of this class of ice-related deposits (e.g. Head et al., 2003, 2010; Milliken et al., 2003; Levy et al., 2010), as well as periods of Late Amazonian glaciation (e.g. Mustard et al., 2001; Head et al., 2003, 2008; Schon et al., 2009; Dickson et al., 2010, 2011).

Based on a survey of targeted CRISM observations over exposed open-basin lake deposits, we have observed that the number of these deposits containing spectrally identified aqueous alteration and evaporite minerals is small (10) compared with the overall number of deposits with CRISM coverage (34). This is a divergence from the common mineralogies observed in terrestrial lacustrine deposits (Jones and Bowser, 1978; Blatt et al., 1980; Leeder, 1999; Wetzel, 2001), and could be due to a spectrally obscuring unit, a lack of spectral coverage at adequate resolution or an unaltered, detrital composition of the sedimentary deposits; however, it appears that neither a spectrally obscuring unit or a lack of data coverage can be the sole cause of this disparity. Based on our observations, we draw the following conclusions:

1. *In situ* aqueous alteration and mineral precipitation was not a dominant process within the majority of open-basin lake systems.
2. The observed aqueous alteration minerals and carbonate in exposed open-basin lake deposits are present as transported material.
3. These materials are reflective of the composition of the bedrock in their watershed.

These conclusions are consistent with the findings of previous workers that suggest aqueous alteration and evaporite minerals identified in individual open-basin lake deposits are present as transported material as opposed to having formed *in situ* (e.g. Ehlmann et al., 2008a, 2008b; Dehouck et al., 2010; Ansan et al., 2011). The lack of evidence for *in situ* alteration and mineral precipitation can be explained by either a water chemistry that was not conducive to such processes, or a short-lived period of lacustrine activity for most open-basin lakes. Determining which of these two hypotheses better explains the observations presented here is very important, as both of these scenarios have strong implications for the climate and hydrologic cycle of Mars during this early period of lacustrine activity.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.icarus.2012.02.027.

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